

Pinball Brief 1 – Ramps

One of the most breathtaking contraptions that continuously fascinates people of all ages is the Rube Goldberg machine. Of the many possible components of a Rube Goldberg machine, perhaps none is more important than the ball rolling down an inclined plane. Almost all implementations of a Rube Goldberg machine will have a rolling ball, such as in *Mouse Trap*. Pinball games use the general audience's love of balls rolling down inclined planes to implement a key feature of many modern pinball tables – the ramp. So, how do pinball manufacturers use ramps and their material composition to enhance gameplay, which types of ramps do players prefer, and how can subtle choices like ramp material affect players' enjoyment of games?

The first pinball machine to use a ramp was the *Nine Sisters* table in 1953. The game features a small and compact, b-shaped metal ramp along the left side of the table that is far different from the long, winding ramps most modern tables contain [10]. Until the 1980s, ramps were a rare inclusion in pinball machines and were generally treated as more of an afterthought by manufacturers. One of the first appearances of a ramp crossing the entire table horizontally is in the 1980 game *Xenon* by Bally. Notably, *Xenon* features a plastic component of its ramp, marking the beginning of a divide in the pinball community on the composition of ramps [2]. Two major ramp materials and designs have emerged: metal wireframe ramps, and plastic clear ramps. Since *Fire!* in 1987 became the first machine to fully integrate ramps into the normal gameplay loop, the debate over the structure of ramps has become increasingly relevant.

Metal ramps are generally composed of two to four long metallic wires. If there are two wires, they are generally placed below the bottom of the ball, and they will be spaced apart such that the distance between the two wires is less than the diameter of the pinball. If there are four wires, they are generally laid out in a circular pattern, with the ball rolling between all four wirings. The most common metal the wiring is made from is Grade 304 stainless steel, which provides a cost-effective and durable way to prevent oxidation of the ramp [7]. Type 304 stainless steel is generally 18% Chromium, providing for corrosion resistance, and 8% nickel, giving the ramp an austenitic structure and reducing magnetism [11]. Grades 302, 304L, and 316 are also used on occasion [4]. Some machines, such as *Lord of the Rings* and *Congo*, implement brass-plated wire ramps for aesthetic reasons.



Multiple wireframes appear in *Avengers: Infinity Quest*. [1]

Due to the ability for plastic to take on many shapes, plastic ramps can have much more varying appearances compared to metal ramps. They can be very wide, sometimes five times the diameter of the ball, or very thin, only being slightly wider than the ball. Almost all plastic ramps

in pinball are transparent or translucent, such as the one in *Xenon*. Plastic ramps are most often composed of PETG (Polyethylene Terephthalate Glycol), and sometimes acrylic/Polymethyl methacrylate or Lexan/Polycarbonate [9]. PETG in ramps became popular in the 1990s in pinball machines manufactured by Williams Electronics, such as *Star Trek: The Next Generation*. PETG is generally chosen as a material for its high impact strength, durability and flexibility, which allows it to outperform other common plastics such as acrylic [12]. Due to its use as a popular filament, PETG, along with PET, also allows pinball enthusiasts to replace or modify their own ramps using a 3D printer [5].

Both plastic and metal ramps have advantages and disadvantages, causing much debate among players. Metal wire ramps are preferred by some players for many sensory reasons, such as their shiny and glossy aesthetic or the “pinging” noise output when the pinball rolls down them. Materially, metal ramps are generally preferred since they are harder to break and require less cleaning [8]. Plastic ramps can become noticeably scratched or covered in dirt and grime due to their translucent nature. Oxidation negatively affects both metal and plastic ramps, with plastic ramps becoming more brittle over time. However, plastic ramps are more easily shaped into unique forms, which can allow them to become more visually ingrained with the pinball machine’s theme.



Both wireframe and plastic ramps appear in *The Mandalorian*. [3]

So which type of ramp material is superior: metal, or plastic? One way to find out is to ask pinball players directly. In an online poll of 330 active contributors to a pinball forum, about 90% said they preferred wire metal ramps over clear plastic ramps [6]. Despite the material advantages and disadvantages that both metal and plastic have, wireform is generally preferred in the pinball community. The psychological and sensory satisfaction of having a metallic pinball smoothly roll through a wire ramp fulfills the desires that many players have for playing pinball in the first place [8].

Despite the advantages metal ramps may have, there are still many great uses for plastic ramps, especially when they fit the theming of the table. For example, *Hot Wheels* uses orange and blue plastic ramps to mimic Hot Wheels racetracks, while *Xenon* integrates its plastic tube-ramp into its sci-fi theme. Some tables integrate both plastic and metal ramps into gameplay. Overall, there are material and theming-related advantages and disadvantages to both plastic and metal ramps, and both have fascinating material compositions. Next time you are at a pinball table, consider how the manufacturer’s choice of ramp material affects your gameplay. More broadly, notice how in any game, seemingly small design choices, like a choice of metal or plastic ramps, can have a large, yet sometimes unseen, impact on immersion.

Works Cited

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